

CS 524: Computational Complexity Theory

Course Code	CS 524
Course Name	Computational Complexity Theory
L-T-P-C	3-1-0-4
Prerequisites	CS-212, CS-304 or instructor's consent
Intended for	M.Tech/Ph.D./B.Tech 3rd/4th year
Distribution	Elective
Mutual Exclusion	-
Approval	59th BoA (Apr 2025)

Course Contents

- **Unit I:** Review: P, NP, coNP, reductions, completeness, Cook-Levin theorem, some NP-complete problems. [2 Lectures]
- **Unit II:** Diagonalization, time hierarchy theorems, Ladner's theorem, Relativization, limits of diagonalization (Baker-Gill-Solovay theorem) [5 Lectures]
- **Unit III:** Space complexity, Savitch's theorem, PSPACE-completeness and NL-completeness, Immerman-Szelepcse nyi theorem, Polynomial hierarchy. [7 Lectures]
- **Unit VI:** Boolean circuits: P/poly, Karp-Lipton Theorem, NC and AC, P-completeness. [5 Lectures]
- **Unit V:** Randomized computation: BPP, RP, coRP, ZPP, Sipser-Gacs theorem, Valiant-Vazirani theorem. [6 Lectures]
- **Unit VI:** Interactive Proofs: AM and MA, graph non-isomorphism, IP = PSPACE, zero-knowledge proofs. [7 Lectures]
- **Unit VII:** Probabilistically Checkable Proofs: PCP theorem, equivalence of two views of the PCP theorem (proof checking and inapproximability), hardness of approximation. [6 Lectures]
- **Unit VIII:** Circuit lower bounds: Hastad's switching lemma, Parity is not in AC⁰. [5 Lectures]

Text Books

1. Sanjeev Arora and Boaz Barak, **Computational Complexity: A Modern Approach**, Cambridge University Press, 2009.
2. Steven Rudich and Avi Wigderson, **Computational Complexity Theory**, American Mathematical Society, 2004.
3. Avi Wigderson, **Mathematics and Computation: A Theory Revolutionizing Technology and Science**, Princeton University Press, 2019.
4. Oded Goldreich, **Computational Complexity: A Conceptual Perspective**, Cambridge University Press, 2008

References

1. Uwe Schöningh, **Gems of Theoretical Computer Science**, Springer, 1998